Cail Daley

I am an observational cosmologist studying the growth of cosmic structure with gravitational lensing, using agentic engineering practices to scale scientific labor and tackle more ambitious problems.

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POSITIONS

2024–PRESENT	**Postdoctoral Researcher**	CEA Paris-Saclay, CosmoStat
2019–2024	**Graduate Research Assistant**	University of Illinois Urbana-Champaign

EDUCATION

2024 — **Ph.D., Astronomy** — University of Illinois Urbana-Champaign
Advisor: Gil Holder.
"CMB Lensing Measurements with Two Years of Data from the SPT-3G Survey."

2018 — **B.A., Astronomy & Physics** — Wesleyan University
Advisor: A. Meredith Hughes. Thesis high honors.
"Using Vertical Structure to Infer the Dynamical Mass Hidden in the AU Mic Debris Disk."

RESEARCH

All current research is done with flocks of AI agents collaborating around the clock.

2024–PRESENT **Cosmic shear with UNIONS**

- Lead postdoc on the UNIONS cosmic shear analysis. Presented the first UNIONS cosmic shear constraints at the Rencontres de Moriond (2026) and drove the B-mode validation ($A\&A$, in press), one of the first observational cosmology papers carried out end to end by AI agents.
- Heading two papers for the next release, targeted for mid-2027: the shape-measurement software and a tomographic shear catalog covering over 5,000 deg^2.

2024–PRESENT **Euclid × CMB lensing**

- Coordinating a cross-correlation project between the Euclid and SPT-3G collaborations that will provide some of the tightest cosmological constraints from either; paper targeted for mid-2027.

2024–PRESENT **Kinematic lensing**

- Supervising master's students on symmetry-based kinematic lensing, now applied to integral-field survey data.

2019–2024 **CMB lensing with SPT-3G**

- Ran one of three independent pipelines in the SPT-3G D1 lensing analysis, cross-validating the deepest lensing maps to date; produced the input maps and point-source masks used by other SPT-3G D1 analyses (3,286 observations, ∼60,000 CPU hours).

2015–2018 **Debris disks**

- Led ALMA observations of the AU Mic debris disk, inferring the mass of stirring bodies from its vertical structure (first author, *ApJ*, 2019).

COLLABORATIONS & ROLES

2024–PRESENT **Euclid Consortium** Member of the CMB cross-correlations science working group; lead of the Euclid × SPT-3G cross-correlation project within the Euclid–SPT-3G memorandum of understanding; global maintainer of the consortium's GitHub organization.

2024–PRESENT **UNIONS** Member of the weak-lensing team; lead author of the shape-measurement software, shear catalog and B-mode systematics papers.

2019–PRESENT **SPT-3G** Collaboration member; one of the main contributors to the SPT-3G D1 lensing analysis.

2025–PRESENT **Lightcone Research** One of four original core members of an open-source effort building tools for verifiable, AI-mediated science, backed by CNRS and UC Berkeley; helped run its developer summit (Berkeley, 2026).

PUBLICATIONS

Two first-author papers, substantive contributions to ten more, and 38 other collaboration papers, chiefly SPT-3G.

h-index 21, ∼2,100 citations (NASA ADS, October 2026)

2026 **C. Daley** et al., "UNIONS-3500 Weak Lensing: II. B-mode validation for cosmic shear." *A&A*, in press.

2019 **C. Daley** et al., "The Mass of Stirring Bodies in the AU Mic Debris Disk Inferred from Resolved Vertical Structure." *ApJ* 875, 87.

AWARDS & FELLOWSHIPS

2026 CNRS AI Rising Talents, finalist one of five from 42 applicants

2023 APS Division of Astrophysics Student Travel Grant

2021 Chambliss Astronomy Achievement Student Award AAS 238th Meeting

2019 Center for Astrophysical Surveys Graduate Fellowship UIUC, $30,000

2015 NASA Connecticut Space Grant Undergraduate Research Fellowship

SUPERVISION

2025–PRESENT **Serena Nakhle** master's, CosmoStat. First application of symmetry-based kinematic lensing to integral-field survey data (SAMI, WiFeS).

2024–2025 **Jordy Ram** master's, CosmoStat. Kinematic lensing methodology and its first extension to elliptical galaxies; paper in preparation.

2022–2023 **Zimo Qu** undergraduate, UIUC. Stellar flares observed simultaneously by SPT and TESS.

2020–2021 **Society for Equity in Astronomy** Mentor to three undergraduate students.

TALKS

NOV 2026 Colloque national Dark Energy, Paris INVITED · FORTHCOMING
Joint UNIONS cosmology talk with C. Murray

SEP 2026 FORTH Workshop on Physics-Informed Machine Learning, Heraklion KEYNOTE & TUTORIAL
"Scaling Scientific Labor with Agentic Engineering"

MAR 2026 Rencontres de Moriond, Cosmology CONTRIBUTED
"UNIONS-3500: First Cosmological Constraints from 2D Cosmic Shear"

FEB 2026 CosmoStat Learning Slot TUTORIAL SERIES
"Agentic AI for Scientific Research"

JUL 2025 TITAN/TOSCA Meeting, European weak-lensing consortium
"UNIONS: First Cosmic Shear Constraints from 3000 deg^2 of Northern Sky"

APR 2025 NRC Herzberg Astronomy & Astrophysics Colloquium, Victoria INVITED
"Cosmology with SPT-3G"

Apr 2025	UBC Cosmo Coffee, Vancouver **invited** "Cosmology with SPT-3G"
Mar 2025	56th ESLAB Symposium, Leiden **keynote** "Cosmology with SPT-3G"
Jan 2025	Deep CosmoStat Days, CEA Paris-Saclay "SPT Component Separation Challenges"
Jan 2024	AAS 243rd Meeting **dissertation talk** "CMB Lensing Measurements with Two Years of Data from the SPT-3G Survey"
Apr 2023	APS April Meeting "CMB Lensing Measurements with SPT-3G"
May 2022	KIPAC Tea Talk, SLAC **invited** "Lensing Maps and Transient Science with the South Pole Telescope"

SERVICE & OUTREACH

Feb 2027	Organiser, SciAI Days, CEA Paris-Saclay departmental workshop on AI for scientific research
2026	Paris Astronomy on Tap public talk, "Astronomy at the End of the World"
2020–2023	SPT First Discoveries science lessons at Fiske Elementary, Chicago
2020–2022	Organiser, UIUC Astronomy on Tap monthly public events
2019	Education Justice Project astronomy and programming workshop at Danville Correctional Center

SKILLS

Software development	Python, Julia, Unix, HPC and SLURM, Snakemake, Apptainer, git and CI, property-based testing, LaTeX, web development.
Cosmology	CAMB, CosmoSIS, CLOE (the Euclid likelihood), GLASS, HEALPix, NaMaster, TreeCorr, lenspyx, emcee; Bayesian inference.
Agents	Claude Code, Codex and Pi; multi-agent orchestration; author of felt, Shuttle and hark; agent-built software in Go, TypeScript and Elixir/Erlang.
Languages	English (native), French (B2).